Southern Cross University

The endocrine system

Introduction, oxytocin, ADH, growth hormone

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Objectives

- Define the 'endocrine system' and 'hormone'
- Give an example of a homeostatic control mechanism involving 'negative feedback'.
- Explain the reason that the hypothalamus is thought to be the controlling organ of the endocrine system.
- Explain the effects of the two hormones secreted from the posterior pituitary gland (oxytocin and ADH) and give an example of when they would be useful as a pharmaceutical agent.
- Explain the actions of growth hormone and the results of excesses and deficiencies during childhood and after puberty.
- Drugs to know: Desmopressin, somatropin, vasopressin, oxytocin

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Medications of the Endocrine System:

Lesson 1

Review of the Endocrine System – Part 1

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Definitions

The **endocrine system:** is a system of glands that secrete hormones directly into the circulatory system. From there, the hormones are often carried to a distant organ.

- The endocrine system is in contrast to the exocrine system, which secretes its hormones using ducts.

Hormones: From the Greek word meaning "impetus"; A biochemical that is produced by glands in the body transported ** by the circulatory system to another organ to regulate physiological and behavioural activities. They serve as a form of communication

**A few hormones act locally as well (oestrogen, testosterone).

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Hormones

- Chemical messengers normally secreted into the bloodstream
- Specific rates and rhythms of secretion
  - Diurnal, pulsatile and cyclic, and patterns depending on circulating substances
- Operate within feedback systems
- Affect only cells with appropriate receptors

The liver inactivates hormones, rendering the hormones more water soluble for renal excretion

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Mechanisms of hormone action

Hormones produce cellular changes in target cells – examples:

- Stimulate mitosis
- Induce secretory activity
- Stimulate protein synthesis
- Activate or deactivate enzyme systems

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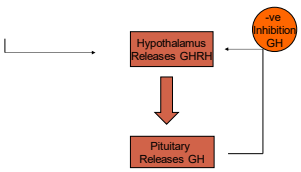
Regulation of hormone release

- Hormones are released:
  - In response to an alteration in the cellular environment
  - To maintain a regulated level of certain substances or other hormones
- Hormones are regulated by chemical, hormonal, or neural factors
  - Negative feedback
  - Positive feedback (rarely)

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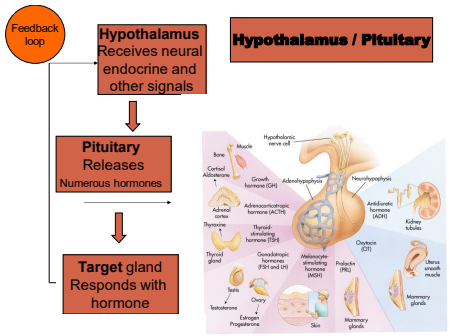
Hormonal Feedback

- Usually negative

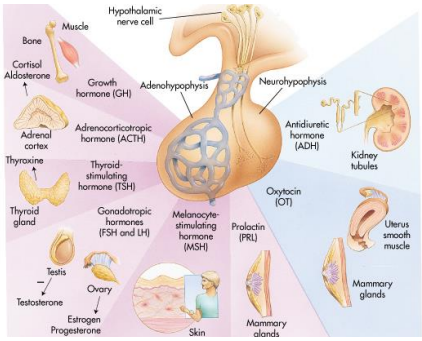


- Seldom positive
  - Example: day 12 of the menstrual cycle, increasing levels of oestrogen from ovary act on hypothalamus to INCREASE release of GRH

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Medications of the Endocrine System:

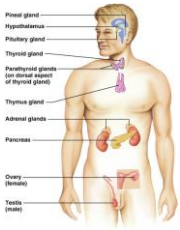
Lesson 2

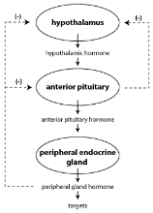
Review of the Endocrine System – Part 2

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Feedback used in the endocrine system

- Most often: negative feedback





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The hypothalamus

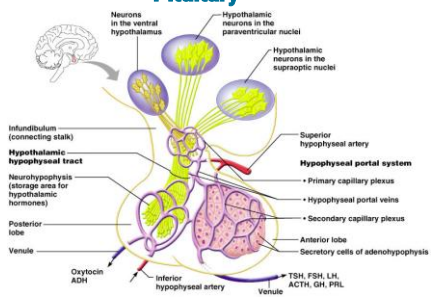
- A neural and an endocrine organ
- Maintains homeostasis
- Responds to most hormones of the body
- Receives neural signals from most other areas of the brain.
- Sends neural information throughout the brain
- Synthesizes and secrets its own hormones
- Inputs:
  - Stress, Pain, Body weight, Temperature, Emotions, Various hormones…



Medical News Today

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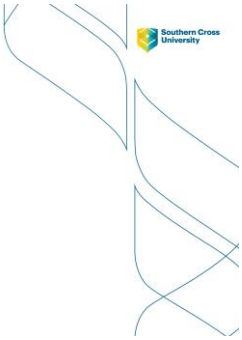
Pituitary



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Posterior pituitary hormones

- Synthesized with their binding proteins in the hypothalamus
- Secreted by the posterior pituitary
  - Antidiuretic hormone (ADH)
    - THE WATER SAVER
    - Controls plasma osmolality
  - Oxytocin
    - Uterine contractions and milk ejection in lactating women



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Medications of the Endocrine System:

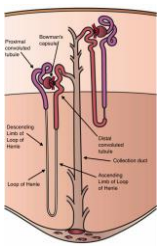
Lesson 3

Antidiuretic hormone (ADH)

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ADH (vasopressin)

- **Too little ADH**, the water saver results in a condition termed ***diabetes insipidus***.
- "Diabetes" comes Ancient Greek διαβήτης (diabētēs) which literally means "a passer through; a siphon"
- "Insipidus" comes from the Latin: *in-* "not" + *sapidus* "tasty" from *sapere* "have a taste"
- Diabetes insipidus is a disease with tasteless urine (in contrast to having sweet urine in diabetes mellitus.) Patients get dehydrated because they cannot save water.



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ADH (vasopressin) replacement: Desmopressin

- Synthetic version of ADH, the water saver (vasopressin)
- Works like ADH
- Main uses include diabetes insipidus but also used in
  - Bed wetting
  - Nocturia
- People using the medication for nocturia should be educated about hyponatremia
- Often, intranasal administration

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Medications of the Endocrine System:
Lesson 4
Oxytocin

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Oxytocin

- Derived from the Greek words meaning 'swift birth'
- The "bonding hormone", the "love hormone"
- Used to initiate labour, control postpartum haemorrhage, (promote breast milk production - no longer in humans)
- Relatively safe when used at recommended doses, and side effects are uncommon
- Has a half-life of about three minutes when given intravenously. Intranasal oxytocin has a duration of at least 2.25 hours
- Replaces the ergot alkaloid **ergometrine** as the first choice in the uses above, but combinations are still used

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Medications of the Endocrine System:
Lesson 5
Growth Hormone

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Growth hormone

- Increases muscle mass through sarcomere hypertrophy
- Promotes lipolysis
- Increases protein synthesis
- Your client needs to be told how dangerous it is to use growth hormone for muscle growth!
  - Stimulates the growth of all internal organs excluding the brain
  - Promotes gluconeogenesis in the liver (increases glucose levels: diabetes!)

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Growth hormone deficiency

- Growth hormone deficiency (GDH) in childhood is commonly idiopathic (no identifiable cause)
- Adult-onset GHD is commonly due to pituitary tumours

- Somatropin is given for growth hormone deficiencies
- Subcutaneous or intramuscular injections

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Growth hormone excess

- Gigantism occurs when the child is still growing.
- Acromegaly occurs after the epiphyseal plates have closed.





Lanreotide is a synthetic analogue of somatostatin, a naturally occurring inhibitory hormone which blocks the release of growth hormone.

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References

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